MANAGEMENT OPERATING REVIEW - RECORD 014

Helioscope Robotics, Inc. - diligence file reference OPERATING_REVIEWS-014

Supply Operating Review for March 2025: purpose and attendance

Data definitions were attached to the packet for the first time, and two functions discovered they were counting differently. Sunil Achterberg, Director of Field Service, presented the operations section. The next session is set for 2025-05-08. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Emil Farkas attended as Director of Manufacturing Engineering. Notes were circulated on 2025-04-09. Capital spend 280 thousand this period. Depot fill 89 percent at Toronto depot. Work in process 37 units, mostly Castings. Sensor head assembly ran 94 percent first pass.

Action item review opened the meeting rather than closing it, which changed the tone of the whole session. The next session is set for 2025-05-08. One function submitted its pre-read late. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Sunil Achterberg, Director of Field Service, presented the operations section. The Supply Operating Review convened for March 2025. Win rate 30 percent, best in Southwest. On-time delivery 86 percent, Brightwater Motion the outlier. Queue depth 25 sites, Cedar Bluff at the front. Attrition 12 percent annualised in Southwest.

Sections with no decision were noted as read, which returned twenty minutes to the constrained functions. Pre-reads went out ahead of the 2025-04-09 session. Emil Farkas attended as Director of Manufacturing Engineering. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Sunil Achterberg, Director of Field Service, presented the operations section. The next session is set for 2025-05-08. Renata Voss owns the Castings recovery. Sensor head assembly ran 94 percent first pass. Marcus Peil and Nadia Oyelaran disagreed on sequencing. Alder & Vance Retail asked for the HS-200 tote handler configuration.

Timekeeping was enforced by the chief of staff rather than by the chair, which worked better than expected. Pre-reads went out ahead of the 2025-04-09 session. Notes were circulated on 2025-04-09. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. The Supply Operating Review convened for March 2025. One function submitted its pre-read late. Win rate 30 percent, best in Southwest. Utilisation 92 percent, Contract crew Northfield fully committed. Days outstanding 47, Havering Logistics the largest balance. Work in process 37 units, mostly Castings.

Bookings against the operating plan

Two of the three regions carried the number this period, and the third missed on a single account that moved its own board meeting. Discount depth peaked on the Marrowfield Grocers multi-site award. Intake closed at 69 units for March 2025. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Signed volume: 69 units, win rate 30 percent. Win rate held at 30 percent on 121 live opportunities. Capital spend 280 thousand this period. Depot fill 89 percent at Toronto depot. Work in process 37 units, mostly Castings. Sensor head assembly ran 94 percent first pass.

Our win rate improved on new logos and weakened on expansions, an inversion of last period that nobody in the room could yet explain. Win rate held at 30 percent on 121 live opportunities. The book took 69 units, against 77 last period. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Signed volume: 69 units, win rate 30 percent. Havering Logistics accounted for the largest single order. Win rate 30 percent, best in Southwest. On-time delivery 86 percent, Brightwater Motion the outlier. Queue depth 25 sites, Cedar Bluff at the front. Attrition 12 percent annualised in Southwest.

Regional coverage remains uneven, with one seller carrying a territory that the coverage model says needs two. Open pursuits number 121, up from 124. Discount depth peaked on the Marrowfield Grocers multi-site award. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Win rate held at 30 percent on 121 live opportunities. Signed volume: 69 units, win rate 30 percent. Renata Voss owns the Castings recovery. Sensor head assembly ran 94 percent first pass. Marcus Peil and Nadia Oyelaran disagreed on sequencing. Alder & Vance Retail asked for the HS-200 tote handler configuration.

Nobody in commercial believes the current forecast discipline is good enough, and the review agreed to inspect deal-level notes rather than roll-ups. Signed volume: 69 units, win rate 30 percent. Havering Logistics accounted for the largest single order. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Intake closed at 69 units for March 2025. Win rate held at 30 percent on 121 live opportunities. Win rate 30 percent, best in Southwest. Utilisation 92 percent, Contract crew Northfield fully committed. Days outstanding 47, Havering Logistics the largest balance. Work in process 37 units, mostly Castings.

The competitive picture shifted, with two losses attributed to a rival that had not appeared in a deal review for three periods. Discount depth peaked on the Marrowfield Grocers multi-site award. Havering Logistics accounted for the largest single order. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. The book took 69 units, against 77 last period. Win rate held at 30 percent on 121 live opportunities. Uptime 97.5 percent, interventions every 347 hours. Days outstanding 47, Havering Logistics the largest balance. Build plan 74, actual 71, Castings short. Kestrel Drive Systems quoted lead times moved again.

Commercial intake for the period came in ahead of the internal plan on unit count and behind it on gross margin, which is the trade the team was asked to make. The book took 69 units, against 77 last period. Signed volume: 69 units, win rate 30 percent. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Havering Logistics accounted for the largest single order. Win rate held at 30 percent on 121 live opportunities. Win rate 30 percent, best in Southwest. Burn 3120 thousand, runway 16 months. Sensor head assembly ran 94 percent first pass. Alder & Vance Retail asked for the HS-200 tote handler configuration.

Contract to go-live decomposition, sites accepted in March 2025

Site	Contract to survey days	Survey to install days	Install to acceptance days	Total elapsed days	Site class
Mesquite	18	33	12	63	greenfield
Ashland Park	18	25	19	62	retrofit
Riverbend	17	27	10	54	greenfield
Kennesaw	17	25	15	57	greenfield
Fort Dodge	17	31	8	56	retrofit
Salinas Valley	17	25	16	58	greenfield
Chandler South	15	23	11	49	greenfield
Elk Grove	21	25	18	64	retrofit
Cedar Bluff	14	25	13	52	greenfield
Norfolk	16	25	20	61	greenfield
Tulsa	18	29	9	56	retrofit
Redlands	16	25	17	58	greenfield
Grand Forks	18	33	12	63	greenfield
Waukegan	18	25	19	62	retrofit
Mesquite (phase two)	17	27	10	54	greenfield
Ashland Park (phase two)	17	25	15	57	greenfield
Riverbend (phase two)	17	31	8	56	retrofit
Kennesaw (phase two)	17	25	16	58	greenfield

Funnel movement and slippage

Late-stage legal review has become a recurring source of slip, and the general counsel asked for the pattern rather than the anecdotes. Two losses went to the same competitor. Conversion into commercial close remains the weak stage. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. The Alder & Vance Retail pursuit re-dated again. Deals leaving the period: 10, most still dated. Capital spend 280 thousand this period. Depot fill 89 percent at Toronto depot. Work in process 37 units, mostly Castings. Sensor head assembly ran 94 percent first pass.

Slippage this period was not random: the deals that moved had a common feature, which is that the customer had no internal project owner. Conversion into commercial close remains the weak stage. Deals leaving the period: 10, most still dated. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Weighted coverage sits near 121 open pursuits. Sunil Achterberg owns the re-qualification pass. Win rate 30 percent, best

in Southwest. On-time delivery 86 percent, Brightwater Motion the outlier. Queue depth 25 sites, Cedar Bluff at the front. Attrition 12 percent annualised in Southwest.

The largest pursuit in the book carries a dependency on the customer finishing a racking project that has already been delayed twice. Two losses went to the same competitor. Slippage totalled 10 opportunities against 8 before. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Deals leaving the period: 10, most still dated. Conversion into commercial close remains the weak stage. Renata Voss owns the Castings recovery. Sensor head assembly ran 94 percent first pass. Marcus Peil and Nadia Oyelaran disagreed on sequencing. Alder & Vance Retail asked for the HS-200 tote handler configuration.

Procurement cycles at the grocery accounts run longer than the model assumes, and the model has not been updated since it was written. The Alder & Vance Retail pursuit re-dated again. Deals leaving the period: 10, most still dated. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Sunil Achterberg owns the re-qualification pass. Slippage totalled 10 opportunities against 8 before. Win rate 30 percent, best in Southwest. Utilisation 92 percent, Contract crew Northfield fully committed. Days outstanding 47, Havering Logistics the largest balance. Work in process 37 units, mostly Castings.

One displacement pursuit has consumed more engineering support hours than the next four combined, with no commitment in return. Conversion into commercial close remains the weak stage. Weighted coverage sits near 121 open pursuits. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Sunil Achterberg owns the re-qualification pass. Deals leaving the period: 10, most still dated. Uptime 97.5 percent, interventions every 347 hours. Days outstanding 47, Havering Logistics the largest balance. Build plan 74, actual 71, Castings short. Kestrel Drive Systems quoted lead times moved again.

The team agreed to stop reporting weighted pipeline in this meeting, on the grounds that nobody had ever acted on the number. Sunil Achterberg owns the re-qualification pass. Slippage totalled 10 opportunities against 8 before. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Deals leaving the period: 10, most still dated. Conversion into commercial close remains the weak stage. Win rate 30 percent, best in Southwest. Burn 3120 thousand, runway 16 months. Sensor head assembly ran 94 percent first pass. Alder & Vance Retail asked for the HS-200 tote handler configuration.

Installation crew loading, March 2025

Crew	Home region	Certified leads	Field engineers	Utilisation %	Weeks committed forward
Crew Alpha	West	2	4	87	13
Crew Bravo	Southwest	2	9	95	5
Crew Charlie	Great Lakes	2	4	90	11
Crew Delta	Southeast	2	6	97	5
Crew Echo	Northeast	1	5	88	14
Crew Foxtrot	Ontario	1	9	93	5
Crew Gulf	Channel and reseller	1	3	86	13
Crew Hotel	West	1	6	94	9
Contract crew Northfield	Southwest	1	4	89	17
Contract crew Verrado	Great Lakes	1	6	96	7
Contract crew Ellingham	Southeast	1	4	91	11
Commissioning bench	Northeast	1	9	98	7
Crew Alpha (night rotation)	Ontario	1	4	87	15
Crew Bravo (night rotation)	Channel and reseller	1	9	95	7
Crew Charlie (night rotation)	West	1	4	90	13
Crew Delta (night rotation)	Southwest	1	6	97	2
Crew Echo (night rotation)	Great Lakes	1	5	88	11
Crew Foxtrot (night rotation)	Southeast	1	9	93	7

Inventory review

The stockroom is holding parts for a configuration that has not been ordered in three quarters, and no one had noticed. Emil Farkas owns the disposition decision. Buffer stock on the drive unit was raised deliberately. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Floor space in staging is now the binding limit. Work in process moved from 34 to 37 units. Capital spend 280 thousand this period. Depot fill 89 percent at Toronto depot. Work in process 37 units, mostly Castings. Sensor head assembly ran 94 percent first pass.

Long-lead buying decisions made two quarters ago are what the stockroom looks like today, for better and worse. Kit shortages were concentrated in low-value parts. Floor space in staging is now the binding limit. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Excess exposure sits in the superseded sensor head. Emil Farkas owns the disposition decision. Win rate 30 percent, best in Southwest. On-time delivery 86 percent, Brightwater Motion the outlier. Queue depth 25 sites, Cedar Bluff at the front. Attrition 12 percent annualised in Southwest.

Work in process climbed because units are being held for a single missing part rather than because the line slowed down. Excess exposure sits in the superseded sensor head. Buffer stock on the drive unit was raised deliberately. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Work in process closed at 37 units. Floor space in staging is now the binding limit. Renata Voss owns the Castings recovery. Sensor head assembly ran 94 percent first pass. Marcus Peil and Nadia Oyelaran disagreed on sequencing. Alder & Vance Retail asked for the HS-200 tote handler configuration.

Excess and obsolete exposure sits almost entirely in the superseded sensor head, and the disposition decision has been deferred twice. Excess exposure sits in the superseded sensor head. Emil Farkas owns the disposition decision. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Buffer stock on the drive unit was raised deliberately. Kit shortages were concentrated in low-value parts. Win rate 30 percent, best in Southwest. Utilisation 92 percent, Contract crew Northfield fully committed. Days outstanding 47, Havering Logistics the largest balance. Work in process 37 units, mostly Castings.

Consignment terms with two suppliers would change the cash picture more than any other inventory action available. Work in process closed at 37 units. Floor space in staging is now the binding limit. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Kit shortages were concentrated in low-value parts. Emil Farkas owns the disposition decision. Uptime 97.5 percent, interventions every 347 hours. Days outstanding 47, Havering Logistics the largest balance. Build plan 74, actual 71, Castings short. Kestrel Drive Systems quoted lead times moved again.

Inventory turns are a poor metric while the company is scaling, so the review looked at coverage weeks by commodity instead. Buffer stock on the drive unit was raised deliberately. Work in process moved from 34 to 37 units. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Excess exposure sits in the superseded sensor head. Emil Farkas owns the disposition decision. Win rate 30 percent, best in Southwest. Burn 3120 thousand, runway 16 months. Sensor head assembly ran 94 percent first pass. Alder & Vance Retail asked for the HS-200 tote handler configuration.

Field reliability review

Reliability engineering split the top failure modes by build vintage and found that one of them is entirely a legacy population problem. Renata Voss leads the joint remediation plan. Mean time between interventions was 347 hours. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Cedar Bluff remains the weakest site on interventions. Havering Logistics measures uptime on a stricter basis. Capital spend 280 thousand this period. Depot fill 89 percent at Toronto depot. Work in process 37 units, mostly Castings. Sensor head assembly ran 94 percent first pass.

Fleet uptime across all live sites improved, and the improvement is broad rather than driven by one recovering account. Havering Logistics measures uptime on a stricter basis. Renata Voss leads the joint remediation plan. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Fleet uptime averaged 97.5 percent. Uptime reached 97.5 percent, from 97.2 percent. Win rate 30 percent, best in Southwest. On-time delivery 86

percent, Brightwater Motion the outlier. Queue depth 25 sites, Cedar Bluff at the front. Attrition 12 percent annualised in Southwest.

Site-level variation in reliability is wider than the fleet average suggests, and the worst site drags the contractual conversation. Uptime reached 97.5 percent, from 97.2 percent. Fleet uptime averaged 97.5 percent. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Cedar Bluff remains the weakest site on interventions. Mean time between interventions was 347 hours. Renata Voss owns the Castings recovery. Sensor head assembly ran 94 percent first pass. Marcus Peil and Nadia Oyelaran disagreed on sequencing. Alder & Vance Retail asked for the HS-200 tote handler configuration.

The remediation plan with the worst-performing account is joint, with obligations written on both sides and a fixed measurement window. Fleet uptime averaged 97.5 percent. Uptime reached 97.5 percent, from 97.2 percent. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Mean time between interventions was 347 hours. Interventions spaced 347 hours apart on the fleet median. Win rate 30 percent, best in Southwest. Utilisation 92 percent, Contract crew Northfield fully committed. Days outstanding 47, Havering Logistics the largest balance. Work in process 37 units, mostly Castings.

Battery degradation is starting to show at the oldest sites, and the replacement schedule needs to become a plan rather than a reaction. Cedar Bluff remains the weakest site on interventions. Havering Logistics measures uptime on a stricter basis. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Interventions spaced 347 hours apart on the fleet median. Mean time between interventions was 347 hours. Uptime 97.5 percent, interventions every 347 hours. Days outstanding 47, Havering Logistics the largest balance. Build plan 74, actual 71, Castings short. Kestrel Drive Systems quoted lead times moved again.

Uptime is measured differently by two customers than by us, and the review chose to adopt the stricter definition internally. Interventions spaced 347 hours apart on the fleet median. Havering Logistics measures uptime on a stricter basis. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Cedar Bluff remains the weakest site on interventions. Renata Voss leads the joint remediation plan. Win rate 30 percent, best in Southwest. Burn 3120 thousand, runway 16 months. Sensor head assembly ran 94 percent first pass. Alder & Vance Retail asked for the HS-200 tote handler configuration.

Build attainment by week and line, March 2025

Week	Product line	Plan units	Actual units	Variance units	Primary constraint
W05	HS-200 tote handler	8	6	-2	fixture failure
W05	HS-400 pallet mover	4	3	-1	plan revision mid-week
W06	HS-200 tote handler	4	3	-1	sensor head rework
W06	HS-400 pallet mover	3	1	-2	operator absence
W07	HS-200 tote handler	5	3	-2	harness shortage
W07	HS-400 pallet mover	3	2	-1	none
W08	HS-200 tote handler	6	4	-2	plan revision mid-week
W08	HS-400 pallet mover	3	2	-1	drive unit shortage
W09	HS-200 tote handler	6	4	-2	operator absence
W09	HS-400 pallet mover	3	2	-1	test cell queue
W10	HS-200 tote handler	3	4	1	none
W10	HS-400 pallet mover	1	3	2	fixture failure
W11	HS-200 tote handler	3	5	2	drive unit shortage
W11	HS-400 pallet mover	2	3	1	sensor head rework
W12	HS-200 tote handler	3	5	2	test cell queue
W12	HS-400 pallet mover	2	3	1	harness shortage
W13	HS-200 tote handler	4	6	2	fixture failure
W13	HS-400 pallet mover	2	3	1	plan revision mid-week

Spares, depots and warranty

Warranty cost per fleet unit is falling, and the fall is real rather than a timing artefact of claim processing. Kennesaw took the planned sensor head replacements. Return material flow remains slow. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Vehicle stock is masking the true fill rate.

Fill rate ran 89 percent against 92 percent. Capital spend 280 thousand this period. Depot fill 89 percent at Toronto depot. Work in process 37 units, mostly Castings. Sensor head assembly ran 94 percent first pass.

Repair-versus-replace decisions are being made in the field without guidance, and the outcomes vary by engineer. Depot fill rate was 89 percent. Fill rate ran 89 percent against 92 percent. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Marcus Peil approved the critical spares increase. Kennesaw took the planned sensor head replacements. Win rate 30 percent, best in Southwest. On-time delivery 86 percent, Brightwater Motion the outlier. Queue depth 25 sites, Cedar Bluff at the front. Attrition 12 percent annualised in Southwest.

Every field incident that waited on a part this period waited on one of four part numbers. Vehicle stock is masking the true fill rate. Return material flow remains slow. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Fill rate ran 89 percent against 92 percent. Depot fill rate was 89 percent. Renata Voss owns the Castings recovery. Sensor head assembly ran 94 percent first pass. Marcus Peil and Nadia Oyelaran disagreed on sequencing. Alder & Vance Retail asked for the HS-200 tote handler configuration.

Engineers are quietly building their own vehicle stock, which means the depot numbers describe a system nobody uses. Depot fill rate was 89 percent. Fill rate ran 89 percent against 92 percent. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Vehicle stock is masking the true fill rate. Warranty cost per fleet unit is falling. Win rate 30 percent, best in Southwest. Utilisation 92 percent, Contract crew Northfield fully committed. Days outstanding 47, Havering Logistics the largest balance. Work in process 37 units, mostly Castings.

Depot placement was reconsidered, because half of the courier spend goes to sites within a day's drive of a hub. Fill rate ran 89 percent against 92 percent. Marcus Peil approved the critical spares increase. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Vehicle stock is masking the true fill rate. Warranty cost per fleet unit is falling. Uptime 97.5 percent, interventions every 347 hours. Days outstanding 47, Havering Logistics the largest balance. Build plan 74, actual 71, Castings short. Kestrel Drive Systems quoted lead times moved again.

The recall-style replacement of affected sensor heads is proceeding on a planned schedule rather than on failure. Marcus Peil approved the critical spares increase. Depot fill rate was 89 percent. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Fill rate ran 89 percent against 92 percent. Return material flow remains slow. Win rate 30 percent, best in Southwest. Burn 3120 thousand, runway 16 months. Sensor head assembly ran 94 percent first pass. Alder & Vance Retail asked for the HS-200 tote handler configuration.

Carried constraints

Constraint under review: drive-unit lead time, at the qualification underway stage as of this session. Brightwater shipped first-article drive units that cleared dimensional inspection but failed the thermal soak on two of eight samples, so qualification carries into the following quarter. Build plan 74, actual 71, Castings short. Burn 3120 thousand, runway 16 months.

Second-source qualification is slow, unglamorous work that removes the only single point of failure left in the drive train. Supplier on-time delivery came in at 86 percent. Sunil Achterberg issued the first supplier scorecards. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. On-time delivery was 86 percent, from 90 percent. Kestrel Drive Systems missed two consecutive commit dates. On-time delivery 86 percent, Brightwater Motion the outlier. Bookings 69 units across 121 pursuits. Uptime 97.5 percent, interventions every 347 hours. Go-live cycle 49 days at Kennesaw.

Component cost movement was mild, and the review spent its time on availability instead because that is what is hurting. Sunil Achterberg issued the first supplier scorecards. Coverage weeks on long-lead parts were extended. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Expedite activity rose with the build plan of 74. Supplier on-time delivery came in at 86 percent. Requisitions open 23, Emil Farkas holding the list. Contract crew Northfield closed Mesquite without a second visit. Burn 3120 thousand, runway 16 months. Capital spend 280 thousand this period.

Assembly cell performance by shift, March 2025

Cell	Shift	Units processed	First-pass yield %	Rework hours	Scrap USD thousands
Frame and drive assembly	first shift	14	97	4	3
Sensor head assembly	first shift	11	89	100	1
Harness and electrical	first shift	14	95	15	1
Final assembly	first shift	14	96	50	1
Functional test	first shift	13	94	25	4
Pack and ship	first shift	13	97	60	1
Frame and drive assembly	second shift	8	94	5	1
Sensor head assembly	second shift	6	87	105	1
Harness and electrical	second shift	8	93	20	1
Final assembly	second shift	8	94	55	1
Functional test	second shift	7	94	10	2
Pack and ship	second shift	7	92	35	1
Frame and drive assembly	weekend cover	2	97	4	1
Sensor head assembly	weekend cover	1	89	100	1
Harness and electrical	weekend cover	2	95	15	1
Final assembly	weekend cover	2	96	50	1
Functional test	weekend cover	2	94	25	1
Pack and ship	weekend cover	2	97	60	1

People and hiring

Open requisitions were re-ranked against the operating plan, and four were closed rather than left to age quietly. Median time to fill was 61 days. The headcount gap is 10 roles. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Open requisitions number 23. Requisitions fell from 27 to 23. Capital spend 280 thousand this period. Depot fill 89 percent at Toronto depot. Work in process 37 units, mostly Castings. Sensor head assembly ran 94 percent first pass.

Manager span of control has grown past what the review considers sustainable in two functions. Time to fill improved from 58 to 61 days. Open requisitions number 23. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Headcount stood at 201 against a plan of 211. Requisitions fell from 27 to 23. Win rate 30 percent, best in Southwest. On-time delivery 86 percent, Brightwater Motion the outlier. Queue depth 25 sites, Cedar Bluff at the front. Attrition 12 percent annualised in Southwest.

A hiring pause was applied selectively, protecting customer-facing roles and holding back internal ones. Headcount stood at 201 against a plan of 211. Requisitions fell from 27 to 23. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Median time to fill was 61 days. Annualised attrition sits at 12 percent. Renata Voss owns the Castings recovery. Sensor head assembly ran 94 percent first pass. Marcus Peil and Nadia Oyelaran disagreed on sequencing. Alder & Vance Retail asked for the HS-200 tote handler configuration.

Engagement survey results split by tenure rather than by function, with the newest cohort least positive about clarity of direction. Open requisitions number 23. Annualised attrition sits at 12 percent. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Requisitions fell from 27 to 23. Median time to fill was 61 days. Win rate 30 percent, best in Southwest. Utilisation 92 percent, Contract crew Northfield fully committed. Days outstanding 47, Havering Logistics the largest balance. Work in process 37 units, mostly Castings.

Time to fill improved for engineering roles and worsened for field roles, which is the opposite of what the plan needed. Open requisitions number 23. Time to fill improved from 58 to 61 days. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. The headcount gap is 10 roles. Median time to fill was 61 days. Uptime 97.5 percent, interventions every 347 hours. Days outstanding 47, Havering Logistics the largest balance. Build plan 74, actual 71, Castings short. Kestrel Drive Systems quoted lead times moved again.

The recruiting funnel converts poorly at the on-site stage for field roles, and the loop design is the likely reason. Median time to fill was 61 days. Open requisitions number 23. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Annualised attrition sits at 12 percent. Headcount stood at 201 against a

plan of 211. Win rate 30 percent, best in Southwest. Burn 3120 thousand, runway 16 months. Sensor head assembly ran 94 percent first pass. Alder & Vance Retail asked for the HS-200 tote handler configuration.

Cash conversion

Nothing in the receivables ledger suggests a credit problem; everything in it suggests a paperwork problem. Two receivables aged past ninety days. Outstanding days improved from 49 to 47. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Receipts of 2810 thousand outpaced invoicing. Nadia Oyelaran moved collections calls to the account teams. Capital spend 280 thousand this period. Depot fill 89 percent at Toronto depot. Work in process 37 units, mostly Castings. Sensor head assembly ran 94 percent first pass.

Terms granted in negotiation are cash decisions made by people who do not carry the cash number. Nadia Oyelaran moved collections calls to the account teams. Outstanding days improved from 49 to 47. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Receipts of 2810 thousand outpaced invoicing. Collections totalled 2810 thousand dollars. Win rate 30 percent, best in Southwest. On-time delivery 86 percent, Brightwater Motion the outlier. Queue depth 25 sites, Cedar Bluff at the front. Attrition 12 percent annualised in Southwest.

Milestone invoicing depends on a signed acceptance document, and the document was the bottleneck rather than the milestone. Nadia Oyelaran moved collections calls to the account teams. Receipts of 2810 thousand outpaced invoicing. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Collections totalled 2810 thousand dollars. Days sales outstanding closed at 47 days. Renata Voss owns the Castings recovery. Sensor head assembly ran 94 percent first pass. Marcus Peil and Nadia Oyelaran disagreed on sequencing. Alder & Vance Retail asked for the HS-200 tote handler configuration.

Deposit terms on new multi-site commitments materially change the working capital profile of the site queue. Two receivables aged past ninety days. Days sales outstanding closed at 47 days. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Nadia Oyelaran moved collections calls to the account teams. Outstanding days improved from 49 to 47. Win rate 30 percent, best in Southwest. Utilisation 92 percent, Contract crew Northfield fully committed. Days outstanding 47, Havering Logistics the largest balance. Work in process 37 units, mostly Castings.

Customer-side accounts payable portals are a real source of delay, and three accounts require a different submission path. Collections totalled 2810 thousand dollars. Two receivables aged past ninety days. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Nadia Oyelaran moved collections calls to the account teams. Marrowfield Grocers requires a separate submission path. Uptime 97.5 percent, interventions every 347 hours. Days outstanding 47, Havering Logistics the largest balance. Build plan 74, actual 71, Castings short. Kestrel Drive Systems quoted lead times moved again.

Progress billing on long deployments would smooth the cash profile, and the commercial team has resisted it for good reasons. Nadia Oyelaran moved collections calls to the account teams. Collections totalled 2810 thousand dollars. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Two receivables aged past ninety days. Outstanding days improved from 49 to 47. Win rate 30 percent, best in Southwest. Burn 3120 thousand, runway 16 months. Sensor head assembly ran 94 percent first pass. Alder & Vance Retail asked for the HS-200 tote handler configuration.

Yield excursion and containment log, March 2025

Week	Cell	First-pass yield %	Units affected	Containment in force	Disposition
W05	Final assembly	92	10	shipments held	return to supplier
W06	Sensor head assembly	99	6	inspection lifted, trigger armed	hold pending study
W07	Harness and electrical	90	7	full inspection reinstated	rework and release
W08	Final assembly	95	8	standard sampling	scrap
W09	Sensor head assembly	88	5	100 percent bond-line inspection	use as is with concession
W10	Harness and electrical	96	10	sample one in ten	return to supplier
W11	Final assembly	91	9	shipments held	hold pending study

Week	Cell	First-pass yield %	Units affected	Containment in force	Disposition
W12	Sensor head assembly	98	7	inspection lifted, trigger armed	rework and release
W13	Harness and electrical	93	6	full inspection reinstated	scrap
W14	Final assembly	99	9	standard sampling	use as is with concession
W15	Sensor head assembly	89	8	100 percent bond-line inspection	return to supplier
W16	Harness and electrical	97	5	sample one in ten	hold pending study
W17	Final assembly	92	10	shipments held	rework and release
W18	Sensor head assembly	99	6	inspection lifted, trigger armed	scrap
W19	Harness and electrical	90	7	full inspection reinstated	use as is with concession
W20	Final assembly	95	8	standard sampling	return to supplier
W21	Sensor head assembly	88	5	100 percent bond-line inspection	hold pending study
W22	Harness and electrical	96	10	sample one in ten	rework and release
W23	Final assembly	91	9	shipments held	scrap
W24	Sensor head assembly	98	7	inspection lifted, trigger armed	use as is with concession

Financial posture

Cutting capital spend buys months now and costs build capacity later, which is a trade this meeting keeps making. Sunil Achterberg holds the trigger list. Runway stands at 16 months on the middle case. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Test-cell capital was deferred again. Spending ran 3120 thousand against 3040 thousand. Capital spend 280 thousand this period. Depot fill 89 percent at Toronto depot. Work in process 37 units, mostly Castings. Sensor head assembly ran 94 percent first pass.

Burn by category matters more than burn in total, because the total has been flat while its composition changed twice. Test-cell capital was deferred again. Runway moved from 17 to 16 months. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Burn for the period was 3120 thousand dollars. Capital spend was 280 thousand dollars. Win rate 30 percent, best in Southwest. On-time delivery 86 percent, Brightwater Motion the outlier. Queue depth 25 sites, Cedar Bluff at the front. Attrition 12 percent annualised in Southwest.

Headcount is the largest lever on burn and the slowest to act, which is why the trigger list is written in advance. Test-cell capital was deferred again. Spending ran 3120 thousand against 3040 thousand. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Capital spend was 280 thousand dollars. Runway stands at 16 months on the middle case. Renata Voss owns the Castings recovery. Sensor head assembly ran 94 percent first pass. Marcus Peil and Nadia Oyelaran disagreed on sequencing. Alder & Vance Retail asked for the HS-200 tote handler configuration.

Contract labour in deployment is the fastest lever available and the one with the most direct customer consequence. Spending ran 3120 thousand against 3040 thousand. Runway moved from 17 to 16 months. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Burn for the period was 3120 thousand dollars. Sunil Achterberg holds the trigger list. Win rate 30 percent, best in Southwest. Utilisation 92 percent, Contract crew Northfield fully committed. Days outstanding 47, Havering Logistics the largest balance. Work in process 37 units, mostly Castings.

Cash discipline this year came two-thirds from collections and one-third from deferred capital, which is a better mix than cutting people. Sunil Achterberg holds the trigger list. Test-cell capital was deferred again. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Runway stands at 16 months on the middle case. Runway moved from 17 to 16 months. Uptime 97.5 percent, interventions every 347 hours. Days outstanding 47, Havering Logistics the largest balance. Build plan 74, actual 71, Castings short. Kestrel Drive Systems quoted lead times moved again.

Runway is not a single number, and presenting it as one has caused more confusion than it has prevented. Runway stands at 16 months on the middle case. Test-cell capital was deferred again. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Capital spend was 280 thousand dollars. Sunil Achterberg

holds the trigger list. Win rate 30 percent, best in Southwest. Burn 3120 thousand, runway 16 months. Sensor head assembly ran 94 percent first pass. Alder & Vance Retail asked for the HS-200 tote handler configuration.

Supplier delivery and lead time, March 2025

Supplier	Commodity	On-time delivery %	Quoted lead weeks	Actual lead weeks	Watch status
Kestrel Drive Systems	Drive units	89	8	12	cleared
Brightwater Motion	Lidar assemblies	84	10	16	second source in qualification
Halverson Optics	Battery packs	91	12	12	on scorecard review
Penrose Castings	Castings	82	14	17	no action
Ironbark Sheet Metal	Sheet metal	87	15	15	watch
Calder Battery Works	Wire harness	80	11	16	escalated
Tessellate Controls	Controllers	88	12	12	cleared
Ryemoor Wire Harness	Caster and wheel sets	83	13	14	second source in qualification
Kestrel Drive Systems	Lidar assemblies	90	14	14	on scorecard review
Brightwater Motion	Battery packs	85	15	17	no action
Halverson Optics	Castings	92	11	11	watch
Penrose Castings	Sheet metal	81	12	16	escalated
Ironbark Sheet Metal	Wire harness	89	13	13	cleared
Calder Battery Works	Controllers	84	14	20	second source in qualification
Tessellate Controls	Caster and wheel sets	91	15	15	on scorecard review
Ryemoor Wire Harness	Drive units	82	11	14	no action

Product checkpoints

Integration variety across warehouse hosts is the tax the roadmap pays every quarter. Adapter work crowded out platform work again. The HS-400 pallet mover absorbed most of the engineering effort. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Emil Farkas brought the decision to stop one feature. Release staging across sites is now standard. Capital spend 280 thousand this period. Depot fill 89 percent at Toronto depot. Work in process 37 units, mostly Castings. Sensor head assembly ran 94 percent first pass.

The warehouse-management adapter absorbed more engineering effort than any other item, and integration variety is the reason. Field data now sets the reliability backlog order. The HS-400 pallet mover absorbed most of the engineering effort. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Release staging across sites is now standard. Two roadmap checkpoints cleared; one slipped. Win rate 30 percent, best in Southwest. On-time delivery 86 percent, Brightwater Motion the outlier. Queue depth 25 sites, Cedar Bluff at the front. Attrition 12 percent annualised in Southwest.

Customer-specific work is crowding out platform work, and the review asked for the ratio to be measured rather than debated. The HS-400 pallet mover absorbed most of the engineering effort. Two roadmap checkpoints cleared; one slipped. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Emil Farkas brought the decision to stop one feature. Adapter work crowded out platform work again. Renata Voss owns the Castings recovery. Sensor head assembly ran 94 percent first pass. Marcus Peil and Nadia Oyelaran disagreed on sequencing. Alder & Vance Retail asked for the HS-200 tote handler configuration.

Product management brought a decision to stop building one feature, and the review supported stopping it. The HS-400 pallet mover absorbed most of the engineering effort. Emil Farkas brought the decision to stop one feature. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Documentation lags the current release. Field data now sets the reliability backlog order. Win rate 30 percent, best in Southwest. Utilisation 92 percent, Contract crew Northfield fully committed. Days outstanding 47, Havering Logistics the largest balance. Work in process 37 units, mostly Castings.

Simulation coverage before release improved, and the last two releases produced fewer field regressions as a result. The HS-400 pallet mover absorbed most of the engineering effort. Field data now sets the reliability backlog order. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Documentation lags the current release. Two roadmap checkpoints cleared; one slipped. Uptime 97.5 percent, interventions every 347

hours. Days outstanding 47, Havering Logistics the largest balance. Build plan 74, actual 71, Castings short. Kestrel Drive Systems quoted lead times moved again.

Stopping work is harder than starting it, and this session stopped one thing on purpose. Emil Farkas brought the decision to stop one feature. The HS-400 pallet mover absorbed most of the engineering effort. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Adapter work crowded out platform work again. Release staging across sites is now standard. Win rate 30 percent, best in Southwest. Burn 3120 thousand, runway 16 months. Sensor head assembly ran 94 percent first pass. Alder & Vance Retail asked for the HS-200 tote handler configuration.

Recorded disagreement

Operations and commercial did not reach agreement on how the site queue should be sequenced, and the chair decided. The disagreement is recorded rather than resolved. Sunil Achterberg and Emil Farkas read the data differently. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. No consensus was reached; the Director of Supply Chain decided. The minority view is preserved in the record. Capital spend 280 thousand this period. Depot fill 89 percent at Toronto depot. Work in process 37 units, mostly Castings. Sensor head assembly ran 94 percent first pass.

One participant asked for the decision to be revisited with better data, and the request was granted with a date attached. The minority view is preserved in the record. No consensus was reached; the Director of Supply Chain decided. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. A further test was added because of the dissent. Nadia Oyelaran asked for the decision to be revisited. Win rate 30 percent, best in Southwest. On-time delivery 86 percent, Brightwater Motion the outlier. Queue depth 25 sites, Cedar Bluff at the front. Attrition 12 percent annualised in Southwest.

A minority view held that the second-source qualification was being rushed, and a further test was added because of it. The disagreement is recorded rather than resolved. The objection was noted against the decision. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. A further test was added because of the dissent. No consensus was reached; the Director of Supply Chain decided. Renata Voss owns the Castings recovery. Sensor head assembly ran 94 percent first pass. Marcus Peil and Nadia Oyelaran disagreed on sequencing. Alder & Vance Retail asked for the HS-200 tote handler configuration.

The plant manager objected to another plan revision inside the month and asked for a freeze window, which was granted in part. Nadia Oyelaran asked for the decision to be revisited. The minority view is preserved in the record. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. The objection was noted against the decision. No consensus was reached; the Director of Supply Chain decided. Win rate 30 percent, best in Southwest. Utilisation 92 percent, Contract crew Northfield fully committed. Days outstanding 47, Havering Logistics the largest balance. Work in process 37 units, mostly Castings.

Whether a deferral saves money depends on which function is counting, and both counts were tabled. No consensus was reached; the Director of Supply Chain decided. The disagreement is recorded rather than resolved. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Sunil Achterberg and Emil Farkas read the data differently. The minority view is preserved in the record. Uptime 97.5 percent, interventions every 347 hours. Days outstanding 47, Havering Logistics the largest balance. Build plan 74, actual 71, Castings short. Kestrel Drive Systems quoted lead times moved again.

Disagreement was recorded rather than smoothed over, because the two positions rest on different readings of the same data. The disagreement is recorded rather than resolved. A further test was added because of the dissent. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. The objection was noted against the decision. Sunil Achterberg and Emil Farkas read the data differently. Win rate 30 percent, best in Southwest. Burn 3120 thousand, runway 16 months. Sensor head assembly ran 94 percent first pass. Alder & Vance Retail asked for the HS-200 tote handler configuration.

Expedite and premium freight log, March 2025

Part number	Supplier	Reason	Freight mode	Added cost USD thousands
4154-5	Tessellate Controls	engineering change cut-in	courier	19
4159-6	Ryemoor Wire Harness	customer date brought forward	expedited road	17
4164-7	Kestrel Drive Systems	build plan pull-in	air	11
4169-8	Brightwater Motion	supplier missed commit date	standard road with premium	21
4174-0	Halverson Optics	yield loss replacement	courier	13
4179-1	Penrose Castings	spares depot backorder	expedited road	15
4184-2	Ironbark Sheet Metal	engineering change cut-in	air	17
4189-3	Calder Battery Works	customer date brought forward	standard road with premium	11
4194-4	Tessellate Controls	build plan pull-in	courier	21
4199-5	Ryemoor Wire Harness	supplier missed commit date	expedited road	19
4204-6	Kestrel Drive Systems	yield loss replacement	air	15
4209-7	Brightwater Motion	spares depot backorder	standard road with premium	13
4214-8	Halverson Optics	engineering change cut-in	courier	19
4219-0	Penrose Castings	customer date brought forward	expedited road	17
4224-1	Ironbark Sheet Metal	build plan pull-in	air	11
4229-2	Calder Battery Works	supplier missed commit date	standard road with premium	21
4234-3	Tessellate Controls	yield loss replacement	courier	13

Decisions, owners and dates recorded in March 2025

Two owners pushed back on dates they had not been consulted about, and the dates were changed in the room. Marcus Peil inherited an item after a role change. One item closed with measured evidence. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Three items carried forward into 2025-05-08. Owners were named and dates set against 2025-05-08. Capital spend 280 thousand this period. Depot fill 89 percent at Toronto depot. Work in process 37 units, mostly Castings. Sensor head assembly ran 94 percent first pass.

Follow-up between meetings will be handled in writing, with the chair reading the updates rather than convening a session. Renata Voss, VP People, took the largest open item. Three items carried forward into 2025-05-08. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. The oldest item dates from two sessions back. One item closed with measured evidence. Win rate 30 percent, best in Southwest. On-time delivery 86 percent, Brightwater Motion the outlier. Queue depth 25 sites, Cedar Bluff at the front. Attrition 12 percent annualised in Southwest.

Shared ownership was abolished on carried items after a quarter in which shared meant nobody. Marcus Peil inherited an item after a role change. Three items carried forward into 2025-05-08. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Renata Voss, VP People, took the largest open item. The oldest item dates from two sessions back. Renata Voss owns the Castings recovery. Sensor head assembly ran 94 percent first pass. Marcus Peil and Nadia Oyelaran disagreed on sequencing. Alder & Vance Retail asked for the HS-200 tote handler configuration.

Escalation out of this forum happens on capital and on customer commitments, and on nothing else. Three items carried forward into 2025-05-08. The oldest item dates from two sessions back. A supplier summit gave unusually current commitments. Suppliers brought their own commitment data. Marcus Peil inherited an item after a role change. Two due dates were changed in the room. Win rate 30 percent, best in Southwest. Utilisation 92 percent, Contract crew Northfield fully committed. Days outstanding 47, Havering Logistics the largest balance. Work in process 37 units, mostly Castings.

Action register as it stood after the March 2025 session

Action	Accountable owner	First raised	Revised due date	State	Note
OR-1204 Qualify Brightwater Motion as second drive-unit source	Emil Farkas	2025-02-05	2025-05-24	Open, date revised	partial
OR-1420 Refresh	Renata Voss	2025-03-31	2025-04-20	In progress	outstanding

Action	Accountable owner	First raised	Revised due date	State	Note
compensation bands for field engineering					
OR-1423 Cost the second-shift ramp against overtime	Marcus Peil	2025-03-27	2025-04-26	Closed	decision recorded
OR-1426 Re-cut the stage definitions in the pipeline report	Marguerite Do	2025-03-23	2025-05-02	Open	outstanding
OR-1429 Reconcile the two uptime definitions with the customer	Owen Castellanos	2025-03-19	2025-05-08	In progress	outstanding
OR-1432 Decide disposition of the superseded sensor head stock	Marcus Peil	2025-03-15	2025-05-14	Closed	report filed
OR-1435 Stage software releases across sites rather than at once	Griffin Tasse	2025-03-11	2025-05-20	Open	outstanding
OR-1438 Set a plan freeze window inside the build month	Emil Farkas	2025-03-07	2025-05-26	In progress	outstanding
OR-1441 Split the reliability backlog by build vintage	Tobias Nkemdirim	2025-03-03	2025-06-01	Closed	measured result attached
OR-1444 Publish the supplier scorecard with weighted criteria	Hollis Bramwell	2025-02-27	2025-06-07	Open	outstanding
OR-1447 Draft deposit terms for multi-site commitments	Ingrid Solberg	2025-02-23	2025-06-13	In progress	outstanding
OR-1450 Bring deal-level notes for the top ten pursuits	Marguerite Do	2025-02-19	2025-06-19	Closed	memo circulated
OR-1453 Rebuild the site cost model separating retrofit from greenfield	Marcus Peil	2025-02-15	2025-06-25	Open	outstanding
OR-1456 Measure onboarding time to productivity for crews	Renata Voss	2025-02-11	2025-07-01	In progress	outstanding
OR-1459 Read the tooling ownership clauses on four part numbers	Hollis Bramwell	2025-02-07	2025-07-07	Closed	memo circulated
OR-1462 Size the buffer stock on long-lead commodities	Hollis Bramwell	2025-02-03	2025-07-13	Open	outstanding